

CAIRO ENGINEERING BIBLE v2.1

Operational Intelligence Architecture
ColomboAI — CAIRO Lab

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0. Philosophy

CAIRO is not an AI agent. It is an AI Operator.

Agents automate workflows.
Operators run systems.

CAIRO is an Operational Intelligence Stack designed to:

- Perceive environments
 - Understand capabilities
 - Plan safely
 - Execute autonomously
 - Learn continuously
 - Govern reliably
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1. Core Brain Layer

ColomboAI-MC-1 (Primary Model)

MC-1 is the central intelligence engine.

Capabilities:

- Language reasoning
- Tool orchestration
- Cross-domain planning
- API abstraction
- Search abstraction
- Skill composition
- Multi-step decision modeling

Upgrades

OSLA (Omnisensory Language Action)

- Multisensory reasoning abstraction
- Unified sensory-state modeling
- Action-conditioned perception

NVIDIA Isaac GROOT N1.6 (VLA)

- Vision-language-action reasoning
- Robotics execution primitives
- Embodied physical planning

NVIDIA Cosmos-Reason2

- Physical common-sense reasoning
 - Vision-language world modeling
 - Failure anticipation
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2. Perception Layer

The Perception Layer answers:

- Where am I?
- What capabilities exist?

- What state am I in?
- What tools are available?
- What risks exist?

Declarative Capability Perception

WebMCP

- Detect MCP-declared capabilities
- Parse tool schemas
- Register deterministic actions

Inferred Capability Perception

Scraping2Tools

- Converts UI into callable tools
- Extracts form logic
- Maps interactive components

SiteMap2Tools

- Builds navigation graph
- Identifies flow transitions
- Detects state boundaries

Contextual Reasoning Perception

MCP-RAG

- Retrieves environment context
- Tool metadata
- SOP documentation
- Historical memory state
- Constraint information

3. Orchestration Layer

Google ADK

- Deterministic task graphs
- Parallel execution
- Stateful execution
- Safe retries
- Constraint injection
- A2UI binding
- Streaming execution

4. Tool Layer

Integrated systems:

- Agent Skills (Native)
 - Self-Learning Skills
 - Composio SDK (100+ integrations)
 - Composio MCP Library
 - Composio Agent Skills
 - OpenClaw Clawhub (cloud adapted)
 - Antigravity Skills Library
 - Google MCP Toolbox for Databases
 - UCP (Universal Commerce Protocol)
 - DeepCode
 - Nanobot
 - OpenCode (Cloud-first + Hybrid)
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5. Memory Layer

MindsDB Integration

- Structured persistence
 - Predictive modeling
 - Cross-session intelligence
 - Embedded ML over memory
 - Long-term behavior optimization
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6. Observability Layer

- Agent Lightning (trace intelligence)
 - Opik (evaluation & optimization)
 - Execution auditing
 - Performance scoring
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7. Governance Layer

- Skill Scanner
 - Shannon (policy graph enforcement)
 - Permission models
 - Enterprise boundaries
 - Risk scoring
 - Override mechanisms
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8. Execution Stack Summary

Brain:

- MC-1 + OSLA
- GROOT N1.6
- Cosmos-Reason2

Perception:

- WebMCP
- Scraping2Tools
- SiteMap2Tools
- MCP-RAG

Orchestration:

- Google ADK
- A2UI

Tool Layer:

- Agent Skills
- Self-learning skills
- Composio integrations
- Clawhub
- Antigravity skills
- MCP DB Toolbox
- DeepCode
- Nanobot
- OpenCode
- UCP

Memory:

- MindsDB

Observability:

- Agent Lightning
- Opik

Governance:

- Skill Scanner
- Shannon

9. Architectural Philosophy

CAIRO is: Not a chatbot.
Not an automation script.
Not a browser wrapper.

It is a multi-layer operational intelligence ecosystem.

It resembles human intelligence: Perception

Reasoning

Memory

Skills

Action

Self-observation

Governance

10. Mission

Build Operational Intelligence.

Deploy AI Operators at planetary scale.

Enable 24/7 AI Employees.

Abstract system complexity for billions.

The future of AI is not interaction. It is operation.

CAIRO ENGINEERING BIBLE DIAGRAM

